

Temporal-Consensus OCR for Accessible Extraction of Static and Scrolling News Headlines

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Abstract—Text in a television lower-third or scrolling ticker is often absent from the audio transcript, yet it carries names, locations, and breaking-news updates. Framewise optical character recognition (OCR) is inadequate when a moving headline is only partly visible per frame. This paper presents a lightweight local pipeline that samples a resolution-independent headline region; evaluates raw, contrast-equalized, and Otsu-binarized hypotheses; filters recognitions by confidence and dictionary-free lexical evidence; and aggregates them over time. Static text is resolved with a quality-weighted medoid, while scrolling text is reconstructed by fuzzy token overlap. A deterministic benchmark generates 50 four-second clips: 40 positives span static and scrolling layouts under four degradation conditions, and 10 are negative controls. Against center-frame OCR, temporal consensus reduces mean character error rate from 12.34% to 2.10% and raises sequence success at normalized similarity 0.80 from 50.0% to 100.0%. The paired gain is 50.0 percentage points (bootstrap 95% CI: 35.0–65.0; exact McNemar $p = 1.91 \times 10^{-6}$). These results validate the temporal mechanism, not readiness for unconstrained broadcasts; real-news, multilingual, and automatic-localization evaluations remain necessary.

Index Terms—optical character recognition, video text recognition, temporal consensus, news ticker, accessibility, reproducible benchmark

I. INTRODUCTION

Broadcast video contains two information channels: the audible program and text rendered into the image. The latter includes lower-thirds, locations, speaker names, scores, market values, weather warnings, and continuously scrolling headlines. Such text is not guaranteed to occur in closed captions or automatic speech recognition output. Recovering it is therefore useful for search, archival, monitoring, and assistive interfaces.

Reading text from video is more difficult than recognizing a scanned page. Characters may be small, blurred by motion, distorted by compression, or superimposed on a changing background. The ICDAR video-text literature identifies illumination change, camera motion, blur, and real-time constraints as persistent problems [1], [2]. More importantly for a ticker, the desired sequence is not present in any one image. A single frame can be recognized perfectly and still be an incomplete answer.

The original course prototype used a fixed pixel crop, Tesseract, a dictionary threshold, and a set of unique strings. It was neither resolution independent nor reproducibly evaluated, and dictionary filtering could reject newsworthy proper nouns. Its reported confusion matrix could not be regenerated. This ground-up revision asks: *How much can lightweight temporal*

aggregation improve headline recovery over single-frame OCR when the text region is known?

The contributions are threefold:

- 1) A CPU-oriented extraction pipeline that separates spatial localization, OCR hypotheses, confidence-aware filtering, static consensus, and scrolling-text assembly.
- 2) A deterministic video generator and paired evaluation protocol covering two motion modes, four controlled degradations, negative controls, ablations, and error-rate metrics.
- 3) An open implementation with tests, machine-readable predictions, regenerated result tables, and a high-contrast browser report with local text-to-speech controls.

This empirical systems paper contributes a reproducible, interpretable temporal layer rather than a new OCR neural architecture.

II. RELATED WORK

A. OCR and Scene Text

Tesseract established a widely used open OCR pipeline built around line finding, adaptive classification, and linguistic processing [3]. Modern scene-text systems couple learned detectors with sequence recognizers. EAST directly predicts oriented word or line geometry [4]; CRAFT models character regions and affinities [5]; and differentiable binarization learns the thresholding operation within a segmentation detector [6]. PP-OCR emphasizes lightweight deployability [7], with later versions revising both detection and recognition components [8]. The implementation evaluated here uses the small PP-OCRv6 recognition model distributed by RapidOCR; PP-OCRv6 is currently documented as a preprint rather than a peer-reviewed result [9], [10].

The present system assumes a user or upstream component has specified a one-line headline band. This reduces CPU cost but prevents direct comparison with general scene-text spotters.

B. Video Text and Temporal Evidence

Nguyen *et al.* used temporal redundancy for video-text recognition on ICDAR-VIDEO and a YouTube dataset [11]. Newer benchmarks evaluate localization, tracking, and complete sequence transcripts [2]; Sequential Transformer models long-term context jointly for detection and tracking [12]. In contrast, this method aggregates text after ROI selection: it is smaller and inspectable but less general.

C. Preprocessing and Synthetic Evaluation

Otsu thresholding optimizes gray-level class separation [13], while adaptive histogram equalization improves local contrast [14]. Synthetic data enables controlled scene-text experiments [15]. Here it is used only for evaluation: it isolates motion and degradation effects but cannot substitute for a real-broadcast corpus.

III. METHOD

A. Input, Sampling, and Region of Interest

Let a video contain frames F_t at source rate r . Frames are sampled at a configurable rate r_s ; the experiment uses $r_s = 2$ frames/s. A region $R = (x, y, w, h) \in [0, 1]^4$ is stored in normalized coordinates and converted to pixels for each frame, eliminating the prototype’s resolution-specific crop. The recognized image hypothesis for preprocessing variant v is

$$I_{t,v} = P_v(R(F_t)), \quad v \in \{\text{raw, CLAHE, Otsu}\}. \quad (1)$$

Each crop is enlarged with bicubic interpolation before line recognition. Because a lower-third can contain more than one line in practice, this one-line assumption is a deliberate scope limit, not a general solution to layout analysis.

B. Recognition and Candidate Quality

For each $I_{t,v}$, the OCR engine returns text $s_{t,v}$ and confidence $c_{t,v} \in [0, 1]$. Unicode is normalized, unsupported symbols are removed, whitespace is collapsed, and comparison is case insensitive. Unlike the original dictionary test, proper nouns and acronyms are not penalized. Candidate quality is

$$q_{t,v} = 0.72c_{t,v} + 0.18a_{t,v} + 0.10 \min(n_{t,v}/8, 1), \quad (2)$$

where $a_{t,v}$ is the proportion of visible characters that are alphanumeric and $n_{t,v}$ is the number of word tokens. A candidate must contain at least three tokens with alphabetic content and confidence at least 0.30. This suppresses isolated labels such as “LIVE” and times such as “8:30 PM” without requiring an English dictionary.

C. Static Consensus

Multiple preprocessing outputs from one frame are reduced to the highest-quality candidate. For a static overlay, the representative string is a quality-weighted medoid. For candidate i ,

$$S_i = q_i + \sum_j q_j \frac{\text{ratio}(s_i, s_j)}{100} + \frac{\min(n_i, 12)}{20}, \quad (3)$$

where ratio is normalized Levenshtein similarity. Selecting $\arg \max_i S_i$ rewards agreement across frames while retaining a complete, high-confidence observation. It avoids character-level voting, which requires uncertain spatial alignment.

D. Scrolling-Ticker Assembly

Mode can be set explicitly. Automatic mode uses near-duplicate support and observed word-count range: a span of at least two with weak global support indicates a ticker; otherwise repeated high-similarity observations indicate static text.

Ticker candidates remain in timestamp order. Near-duplicate adjacent observations are removed. For two windows $A = (a_1, \dots, a_m)$ and $B = (b_1, \dots, b_n)$, the algorithm searches from the largest possible suffix–prefix overlap downward. An overlap of length k is accepted when the mean fuzzy similarity of $(a_{m-k+1}, \dots, a_m)$ and $(b_1, \dots, b_k)$ is at least 76. Within the overlap, the longer of two near-matching tokens is retained; this repairs boundary fragments such as “OUNCIL”/“COUNCIL” or “PL”/“PLAN.” A character-level overlap of more than five characters is the fallback. Competing chains are ranked first by recovered word count and then by accumulated quality.

Figure 2 shows why temporal assembly is necessary: different frames expose different, overlapping portions of one sentence.

E. Accessible Output

UTF-8 JSON records the transcript, mode, timing, configuration, and retained candidates. An optional high-contrast HTML view offers scalable type, keyboard-visible controls, and local browser speech synthesis. The visible transcript remains inspectable before playback.

IV. EXPERIMENTAL DESIGN

A. Controlled Benchmark

The generator creates 40 positive clips from five neutral, fictional headline strings crossed with two motion modes and four conditions. Each MJPG clip is 640×360, six frames/s, and 24 frames (4 s). Static text is centered and entirely visible. Ticker text is rendered larger than the 600-pixel band and moves right-to-left, so no single crop is guaranteed to contain the complete sentence. Conditions are: (1) clean; (2) 30-level low contrast; (3) Gaussian noise with standard deviation 28; and (4) seven-pixel horizontal blur followed by JPEG quality 25. Ten negative clips contain an empty band or a non-headline label. The random seed, ground truth, and every rendering parameter are stored in the manifest.

This intentionally small dataset is a mechanism test: motion, visibility, and degradation vary while the transcript is known exactly. It does not estimate performance across broadcast networks, languages, fonts, or graphics packages.

B. Ablations and Metrics

Three inference conditions use the same sampled video:

- 1) **Center frame**: raw OCR from the sampled frame nearest the clip midpoint.
- 2) **Center + multiview**: the highest-quality raw, CLAHE, or Otsu hypothesis at that frame.
- 3) **Temporal**: automatic static/ticker inference and aggregation over all samples.

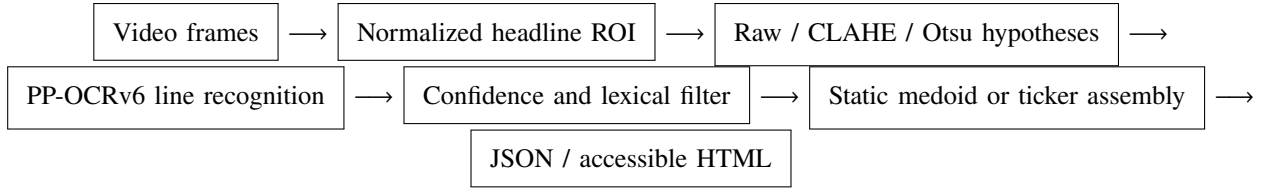


Fig. 1. Pipeline boundary. The system recognizes a supplied single-line region rather than performing unconstrained full-frame text detection.



Fig. 2. Three sampled regions from a synthetic ticker under horizontal blur and JPEG compression. No displayed crop contains the full reference, “RESEARCHERS RELEASE OPEN CLIMATE DATASET,” but chronological overlaps provide the missing context.

For positive clips, character error rate (CER) and word error rate (WER) are Levenshtein distance divided by reference characters and words, respectively. Exact-match rate is also reported. A clip is a sequence-level success when normalized character similarity is at least 0.80. Negative clips make precision measurable; positive clips determine recall. The threshold is declared before comparison and is not an exact-transcription claim.

The success-rate difference is paired by clip. A seeded 5,000-resample paired bootstrap gives a 95% percentile interval [16]; an exact two-sided McNemar test evaluates the discordant binary outcomes. These statistics describe this fixed controlled corpus only.

C. Reproducibility Environment

Experiments ran on Windows 11 and a 12th-generation Intel Core i5-1235U CPU using Python 3.13.9, RapidOCR 3.9.2, ONNX Runtime 1.29.0, OpenCV 5.0.0.93, and NumPy 2.5.2. Inference was CPU-only and used the packaged PP-OCRv6 small recognizer. Generated videos are excluded from version control; one command recreates them and the CSV, JSON, and $\LaTeX$ result artifacts.

V. RESULTS

Table I shows that preprocessing selection alone does not improve center-frame OCR: CER is unchanged (12.34% versus 12.34%) and WER is slightly worse (26.33% versus 26.83%). Thus the temporal gain cannot be attributed to preprocessing.

Temporal consensus reduces CER by 10.24 percentage points and WER by 16.66 points. Exact match rises from 37.5% to 70.0%. All 40 positive clips reach the declared 0.80 similarity threshold, and all 10 controls are rejected, producing 100.00% F1. The paired success gain is 50.0 points (95% bootstrap CI 35.0–65.0); the exact McNemar result is $p = 1.91 \times 10^{-6}$.

Table II locates the effect. Static clips are easy for one well-cropped frame, while all sequence-level gain comes from scrolling clips whose center frame is structurally incomplete. This supports the mechanism, not a claim that temporal aggregation universally improves OCR.

Across conditions, temporal CER is 1.73% for clean, 1.97% for low contrast, 1.73% for Gaussian noise, and 2.98% for blur plus compression. The blur condition is hardest, but every output remains above the 0.80 success threshold. Median end-to-end processing time is 0.62 s per 4-s clip after model initialization, or approximately 0.16 $\times$ video duration on this machine. This is an offline throughput observation, not an interactive latency guarantee.

VI. ERROR ANALYSIS AND DISCUSSION

Twelve temporal outputs are non-exact despite 40/40 threshold successes. Errors fall into three classes. First, blur causes character deletion or duplication, for example “APPROVES” becoming “PPROVES.” Second, a static crop can lose a word boundary (“TEAMADVANCES”). Third, fuzzy ticker reconciliation can preserve a plausible but wrong fragment, such as “COM CLITY” for “COMMUNITY.” The longest headline is occasionally truncated to “CHAMPIONSHIP F.” These examples explain why 100% threshold success must not be described as 100% transcription accuracy; exact match is 70% and WER is 9.67%.

The local pipeline preserves proper nouns, exposes candidates for audit, and separates OCR from temporal logic. Its normalized ROI spans resolutions, and ticker assembly recovers information that no selected frame contains.

The principal limitation is external validity. The benchmark uses five English strings and one OpenCV font family on synthetic backgrounds; degradations approximate, rather than reproduce, broadcast encoders and graphics. The ROI is supplied, text is horizontal and single-line, and the recognizer’s pretraining data are outside experimental control. No comparison is made with cloud OCR, large vision-language models, or end-to-end video text spotters because those systems solve broader tasks and would require a real annotated dataset. Confidence is not calibrated, and the heuristic auto-mode boundary was evaluated on the same controlled design in which it was developed. Finally, the accessible interface has not undergone a user study with blind or low-vision participants.

TABLE I
OVERALL CONTROLLED-BENCHMARK RESULTS. LOWER IS BETTER FOR CER/WER; HIGHER IS BETTER OTHERWISE. PRECISION, RECALL, AND F1 USE THE 0.80 SEQUENCE-SUCCESS RULE.

Method	CER ↓	WER ↓	Exact ↑	Success ↑	Precision	Recall	F1
Center frame	12.34%	26.33%	37.5%	50.0%	100.00%	50.00%	66.67%
Center + multiview	12.34%	26.83%	37.5%	50.0%	100.00%	50.00%	66.67%
Temporal consensus	2.10%	9.67%	70.0%	100.0%	100.00%	100.00%	100.00%

TABLE II
POSITIVE-CLIP RESULTS STRATIFIED BY MOTION MODE.

Mode	Method	CER	Exact	Success
Static	Center frame	0.87%	75%	100%
Static	Temporal	0.75%	80%	100%
Ticker	Center frame	23.80%	0%	0%
Ticker	Temporal	3.45%	60%	100%

Errors in names, numbers, or emergency messages can change meaning. Output should therefore be presented as machine transcription with visible context and confidence, not an authoritative news feed. Future work should use a licensed broadcast corpus, add multi-line detection and tracking, evaluate multilingual recognition, calibrate uncertainty, and conduct an accessibility study.

VII. CONCLUSION

This work replaces an unreproducible framewise prototype with a tested, local, auditable system. On a controlled 50-clip benchmark, temporal overlap reconstructs scrolling sentences that center-frame OCR cannot fully observe, reducing CER from 12.34% to 2.10% while exact accuracy remains 70.0%. The evidence supports effectiveness for known, single-line ticker regions under the tested degradations; broader claims require real-world, multilingual, independently annotated evaluation.

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